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中正中學

CHUNG CHENG HIGH SCHOOL (MAIN)

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PRELIMINARY EXAMINATION 2026
SECONDARY 4
(MARKING SCHEME)

ADDITIONAL MATHEMATICS

4049/02

Paper 2

Wednesday 26 August 2026

Candidates answer on the Question Paper.

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

For Examiner's Use	
Question Number	Marks Obtained
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total Marks	

This document consists of **20** printed pages.

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1) \dots (n-r+1)}{r!}$.

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} ab \sin C$$

- 1 Scientists monitoring a shrinking glacier estimate that its area, $A \text{ km}^2$, is modelled by $A = 850e^{-kt}$, where t is the number of years after the year 2010 and k is a positive constant. In 2025, the area of the glacier was 620 km^2 .

(a) Find the value of k , correct to 4 significant figures.

[3]

Solution	Description	Marks
When $A = 620$, $t = 15$. $620 = 850e^{-15k}$ $e^{-15k} = \frac{62}{85}$ $-15k = \ln\left(\frac{62}{85}\right)$ $k = -\frac{1}{15} \ln\left(\frac{62}{85}\right)$ $= 0.0210344\dots$ $= \mathbf{0.02103}$ (4 s.f.)	substitution take 'ln' on both sides or convert to log form	M1 M1 f.t. A1

- (b) A journalist reported: "At this rate, the glacier would have completely disappeared by the year 2110." Do you agree with the journalist's statement? Use calculations to justify your answer.

[2]

Solution	Description	Marks
When $t = 100$, $A = 850e^{-100\left(-\frac{1}{15} \ln\left(\frac{62}{85}\right)\right)}$ $= \mathbf{104}$ (3 s.f.) Since $A \approx 104 \text{ km}^2 > 0$ at $t = 100$, the glacier has not completely disappeared. The journalist's claim is incorrect .	substitute $t = 100$	M1 f.t. A1

- (c) Give a mathematical reason why the model may not be reliable to determine if the glacier will completely disappear.

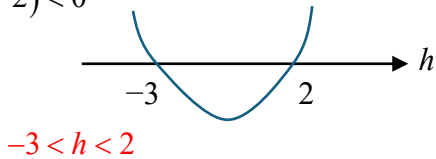
[1]

Solution	Description	Marks
Since $e^{-kt} > 0$, then $850e^{-kt} > 0$. The model predicts $A \rightarrow 0$ but never reaches 0, so it cannot model the glacier fully disappearing.		B1

- 2 (a) Show that the equation $(a^2 - bc)x^2 + 2(b + c)x = 4$ has real roots for all real values of a , b and c . [4]

Solution	Description	Marks
$(a^2 - bc)x^2 + 2(b + c)x - 4 = 0$ Discriminant $= [2(b + c)]^2 - 4(a^2 - bc)(-4)$ $= 4(b^2 + 2bc + c^2) + 16(a^2 - bc)$ $= 4(b^2 + 2bc + c^2 + 4a^2 - 4bc)$ $= 4(b^2 - 2bc + c^2 + 4a^2)$ $= 4[(b - c)^2 + 4a^2]$ $= 4(b - c)^2 + 16a^2$ Since $(b - c)^2 \geq 0$ and $a^2 \geq 0$, $\therefore 4(b - c)^2 + 16a^2 \geq 0$ for all real values of a , b and c . $\therefore$ discriminant ≥ 0 Thus, $(a^2 - bc)x^2 + 2(b + c)x = 4$ has real roots for all real values of a , b and c .	equate to 0 find discriminant expand and simplify either $(b - c)^2 \geq 0$ or $a^2 \geq 0$. Conclude discriminant ≥ 0	M1 M1 M1 f.t. A1
<u>Alternative:</u> Discriminant $= [2(b + c)]^2 - 4(a^2 - bc)(-4)$ $= 4b^2 + 8bc + 4c^2 + 16a^2 - 16bc$ $= 4b^2 - 8bc + 4c^2 + 16a^2$ $= (2b - 2c)^2 + 16a^2$ Since $(2b - 2c)^2 \geq 0$ and $a^2 \geq 0$, $\therefore (2b - 2c)^2 + 16a^2 \geq 0$ for all real values of a , b and c . $\therefore$ discriminant ≥ 0 Thus, $(a^2 - bc)x^2 + 2(b + c)x = 4$ has real roots for all real values of a , b and c .		

- (b) Find the range of values for the constant h for which the curve $y = x^2 - 3hx + 7 - h$ lies above the line $y = 1 - hx$. [4]

Solution	Description	Marks
$y = x^2 - 3hx + 7 - h \quad \dots(1)$ $y = 1 - hx \quad \dots(2)$ Sub. (2) into (1): $x^2 - 3hx + 7 - h = 1 - hx$ $x^2 - 2hx + 6 - h = 0$ $a = 1, b = -2h, c = 6 - h$ For curve lies above line, Discriminant < 0 $(-2h)^2 - 4(1)(6 - h) < 0$ $4h^2 + 4h - 24 < 0$ $h^2 + h - 6 < 0$ $(h + 3)(h - 2) < 0$  $-3 < h < 2$	substitution sub. and let discriminant < 0 factorise	B1 M1 f.t M1 f.t. A1
Alternative: Since the curve lies above the line, $x^2 - 3hx + 7 - h > 1 - hx$ $x^2 - 2hx + 6 - h > 0$ Discriminant < 0 $(-2h)^2 - 4(1)(6 - h) < 0$ $4h^2 + 4h - 24 < 0$ $h^2 + h - 6 < 0$ $(h + 3)(h - 2) < 0$ $-3 < h < 2$		

- 3 (a) Solve the equation $\log_3(1+2x) + 2\log_9(3x-1) = \log_6 1$.

[5]

Solution	Description	Marks
$\log_3(1+2x) + 2\log_9(3x-1) = \log_6 1$ $\log_3(1+2x) + 2\left(\frac{\log_3(3x-1)}{\log_3 9}\right) = 0$ $\log_3(1+2x) + 2\left(\frac{\log_3(3x-1)}{2}\right) = 0$ $\log_3(1+2x) + \log_3(3x-1) = 0$ $\log_3(1+2x)(3x-1) = 0$ $(1+2x)(3x-1) = 3^0$ $6x^2 + x - 1 = 1$ $6x^2 + x - 2 = 0$ $(2x-1)(3x+2) = 0$ $x = \frac{1}{2} \text{ or } -\frac{2}{3} \text{ (rej.)}$	change-of-base $\log_6 1 = 0$ either laws of log convert to exp form	B1 B1 M1 f.t M1 f.t. A1

- (b) Solve the equation $5^x - 4(5^{-x}) = 3$, leaving your answer correct to 2 significant figures. [5]

Solution	Description	Marks
$5^x - 4(5^{-x}) = 3$ Let $u = 5^x$. $u - \frac{4}{u} = 3$ $u^2 - 3u - 4 = 0$ $(u - 4)(u + 1) = 0$ $u = 4$ or $u = -1$ $5^x = 4$ or $5^x = -1$ (rej. $\because 5^x > 0$) $5^x = 4$ $\ln 5^x = \ln 4$ $x = \frac{\ln 4}{\ln 5}$ $= 0.8613...$ $= 0.86$ (2 s.f.)	substitution to form quadratic eq. factorise o.e. reject -ve root take 'ln' or 'lg' on both side	B1 M1 f.t A1 M1 f.t. A1

- 4 (a) A curve has equation $y = \ln\left(\frac{3x+2}{\sqrt{x+5}}\right)$, where $x > 0$.

- (i) Find the gradient of the curve at the point where $x = 1$.

[3]

Solution	Description	Marks
$y = \ln\left(\frac{3x+2}{\sqrt{x+5}}\right)$ $= \ln(3x+2) - \frac{1}{2}\ln(x+5)$ $\frac{dy}{dx} = \frac{3}{3x+2} - \frac{1}{2}\left(\frac{1}{x+5}\right)$ When $x = 1$, $\text{gradient} = \left.\frac{dy}{dx}\right _{x=1}$ $= \frac{3}{3(1)+2} - \frac{1}{2}\left(\frac{1}{1+5}\right)$ $= \frac{31}{60} \text{ or } 0.517 \text{ (3 s.f.)}$	differentiation (either correct or quotient law) substitute $x = 1$ into $\frac{dy}{dx}$	M1 M1 f.t. A1

- (ii) Explain why the curve is strictly increasing for all $x > 0$.

[2]

Solution	Description	Marks
$\frac{dy}{dx} = \frac{6(x+5) - (3x+2)}{2(3x+2)(x+5)}$ $= \frac{3x+28}{2(3x+2)(x+5)}$ For $x > 0$, $3x+28 > 0 \text{ and } 2(3x+2)(x+5) > 0.$ Hence, $\frac{dy}{dx} > 0$. $\therefore$ The curve is strictly increasing for all $x > 0$.	numerator > 0 and denominator > 0 conclude	M1 f.t. A1

- (b) The curve $y = f(x)$ is such that $f'(x) = 4 \sin 2x \cos 2x$. The curve passes through the point $\left(\frac{\pi}{8}, 1\right)$. Find an expression for $f(x)$. [4]

Solution	Description	Marks
$f'(x) = 4 \sin 2x \cos 2x$ $= 2 \sin 4x$	simplify	B1
$f(x) = \int 2 \sin 4x \, dx$ $= -\frac{1}{2} \cos 4x + c$, where c is a constant.	attempt to integrate	M1 f.t
Sub. $\left(\frac{\pi}{8}, 1\right)$ into $f(x)$: $1 = -\frac{1}{2} \cos 4\left(\frac{\pi}{8}\right) + c$ $c = 1$	substitution to find c into their $f(x)$	M1 f.t.
$f(x) = -\frac{1}{2} \cos 4x + 1$		A1

5 A curve has equation $y = 2e^x + 3e^{-2x}$.

- (a) Find $\frac{dy}{dx}$ and hence show that the x -coordinate of the stationary point satisfies the equation $e^{3x} = 3$. [3]

Solution	Description	Marks
$\frac{dy}{dx} = 2e^x - 6e^{-2x}$	differentiate either	B1
At stationary point, $\frac{dy}{dx} = 0$.		
$2e^x - 6e^{-2x} = 0$	let $\frac{dy}{dx} = 0$.	M1 f.t.
$2e^x - \frac{6}{(e^x)^2} = 0$		
$2e^{3x} - 6 = 0$	(steps shown)	
$2e^{3x} = 6$	simplify	A1
$e^{3x} = 3$ (shown)		

- (b) Without solving for the value of x at the stationary point, determine the nature of the stationary point. [3]

Solution	Description	Marks
$\frac{dy}{dx} = 2e^x - 6e^{-2x}$		
$\frac{d^2y}{dx^2} = 2e^x + 12e^{-2x}$	differentiate their $\frac{dy}{dx}$	M1
Since $e^x > 0$ and $e^{-2x} > 0$, $\frac{d^2y}{dx^2} > 0$.		M1 f.t.
$\therefore$ The stationary point is a <u>minimum point</u> .	exp terms > 0	A1

- 6 The polynomial f is given by $f(x) = x^3 + (k+1)x^2 + 3kx + 2k$, where k is constant.

It is also given that $f(x) = (x+1)(x^2 + px + q)$, where p and q are constants.

- (a) Express p and q in terms of k .

[3]

Solution	Description	Marks
$x^3 + (k+1)x^2 + 3kx + 2k = (x+1)(x^2 + px + q)$ $= x^3 + (p+1)x^2 + (p+q)x + q$ By inspection (comparing constants), $q = 2k$ Comparing coefficients of x^2, $k+1 = 1+p$ $p = k$	expansion or comparing coefficients	B1 M1 f.t. A1

- (b) Find the values of k for which $f(x) = 0$ has exactly two real roots.

[3]

Solution	Description	Marks
For $f(x) = 0$ to have exactly 2 real roots, i.e. $x^2 + px + q = 0$ will only have 1 real root which means its discriminant = 0. $p^2 - 4q = 0$ $k^2 - 4(2k) = 0$ $k(k-8) = 0$ $k = 0 \text{ or } 8$	discriminant = 0 find discriminant	B1 M1 f.t. A1

- 7 A circle, with centre C , has an equation $x^2 + y^2 - 8x - 6y + 12 = 0$.

(a) Find the coordinates of C and the radius of the circle.

[3]

Solution	Description	Marks
$x^2 + y^2 - 8x - 6y + 12 = 0$ Comparing, $2g = -8, \quad 2f = -6, \quad c = 12$ Centre $C = (4, 3)$ radius $= \sqrt{(-4)^2 + (-3)^2 - 12}$ $= \sqrt{13}$ units or 3.61 (3 s.f.)	formula	B1 M1 f.t. A1
<u>Alternative:</u> $x^2 + y^2 - 8x - 6y + 12 = 0$ $(x - 4)^2 - 16 + (y - 3)^2 - 9 + 12 = 0$ $(x - 4)^2 + (y - 3)^2 = 13$ Centre $= (4, 3)$, radius $= \sqrt{13}$ units	complete-the-square for either	M1 f.t. A1, A1

The point $R(6, 6)$ lies on the circle.

- (b) Verify that R lies on the line $2x + 3y = 30$ and show that this line is tangent to the circle at R .

[3]

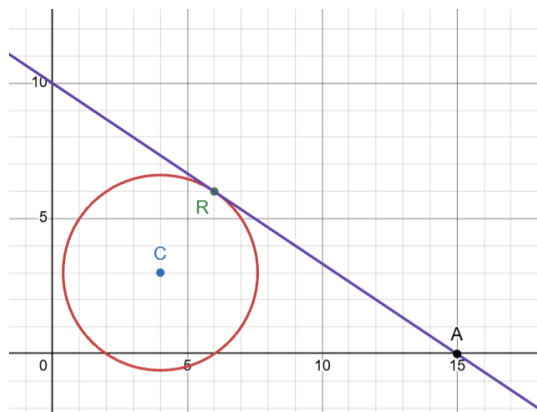
Solution	Description	Marks
Sub. $x = 6$ into $2x + 3y = 30$: $2(6) + 3y = 30$ $3y = 18$ $y = 6$ $\therefore$ Since $R(6, 6)$ satisfies the equation $2x + 3y = 30$, the point R lies on the line.	sub.x- or y-value into line and explain	B1
$m_{\text{radius}} = \frac{6-3}{6-4}$ $= \frac{3}{2}$ and $m_{\text{line}} = -\frac{2}{3}$	find gradient of radius	M1 f.t.
Since $m_{\text{radius}} \times m_{\text{line}} = \frac{3}{2} \times -\frac{2}{3} = -1$, the line is perpendicular to the radius. By property of tangent $\perp$ radius , the line is perpendicular to the circle at R .	tan $\perp$ rad property	A1

<u>Alternative:</u> $2x + 3y = 30$ $y = 10 - \frac{2}{3}x \quad \dots(1)$ $x^2 + y^2 - 8x - 6y + 12 = 0 \quad \dots(2)$ Sub. (1) into (2): $x^2 + \left(10 - \frac{2}{3}x\right)^2 - 8x - 6\left(10 - \frac{2}{3}x\right) + 12 = 0$ $x^2 + 100 - \frac{40}{3}x + \frac{4}{9}x^2 - 8x - 60 + 4x + 12 = 0$ $\frac{13}{9}x^2 - \frac{52}{3}x + 52 = 0$ $13x^2 - 156x + 468 = 0$ $x^2 - 12x + 36 = 0$ $(x - 6)^2 = 0$ $x = 6 \text{ (repeated)}$ Since there is <u>only 1 solution</u>, the line is tangent to the circle at R. <u>OR</u> $\text{discriminant} = (-12)^2 - 4(1)(36)$ $= 0$ Since <u>discriminant = 0</u>, the line is tangent to the circle at R.	substitution solve for x find discriminant and conclude	M1 A1
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The tangent at R meets the x -axis at A .

- (c) Find the area of triangle ACR .

[3]



Solution	Description	Marks
Sub. $y = 0$ into $2x + 3y = 30$: $2x + 3(0) = 30$ $2x = 30$ $x = 15$ $\therefore A(15, 0)$ $\text{Area of } ACR = \frac{1}{2} \begin{vmatrix} 4 & 15 & 6 & 4 \\ 3 & 0 & 6 & 3 \end{vmatrix}$ $= \frac{1}{2} [(0 + 90 + 18) - (24 + 0 + 45)]$ $= \frac{1}{2} (108 - 69)$ $= 19.5 \text{ units}^2$	find A sub. into formula and expand	B1 M1 f.t. A1
<u>Alternative:</u> $AR = \sqrt{(15 - 6)^2 + (0 - 6)^2}$ $= \sqrt{117} \text{ units}^2$ $\text{Area of } ACR = \frac{1}{2} \times CR \times AR$ $= \frac{1}{2} \times \sqrt{13} \times \sqrt{117}$ $= 19.5 \text{ units}^2$	distance AR	M1 f.t. A1

- 8 A particle moves in a straight line and passes through a fixed point O with a velocity of 10 m/s. The acceleration, $a \text{ m/s}^2$, of the particle t seconds after passing O , is given by $a = -3(t+2)^2 + 9$.

- (a) Explain, by showing your working clearly, why the particle only comes to instantaneous rest once in the interval $0 \leq t \leq 3$. [6]

Solution	Description	Marks
$v = \int -3(t+2)^2 + 0 \, dt$ $= -3 \left[\frac{(t+2)^3}{3} \right] + 9t + c$ $= -(t+2)^3 + 9t + c, \text{ where } c \text{ is a constant.}$ When $t = 0$, $v = 10$. $10 = -(0+2)^3 + 9(0) + c$ $10 = -8 + c$ $c = 18$ $\therefore v = -(t+2)^3 + 9t + 18 \text{ or } -t^3 - 6t^2 - 3t + 10$ At instantaneous rest, $v = 0$. $-(t+2)^3 + 9t + 18 = 0$ Let $t = 1$, Refer to alternative method $v = -(1+2)^3 + 9(1) + 18$ $= -27 + 9 + 18$ $= 0$ Since there is no remainder, by Factor Theorem, $(t-1)$ is a factor of v. $-(t+2)^3 + 9t + 18 = (t-1)(-t^2 + ht + k)$ Comparing, $\begin{aligned} -8 + 18 &= -k & \text{and} & & -6 &= 1 + h \\ k &= -10 & & & h &= -7 \end{aligned}$ $v = 0$ $(t-1)(-t^2 - 7t - 10) = 0$ $(t-1)(t+2)(t+5) = 0$ $t = 1 \text{ or } t = -2 \text{ or } t = -5$ Since $t \geq 0$, $t = -2$ and -5 is rejected.	integrate either sub. $t = 0$, $v = 10$ to find c Let $v = 0$ find first factor using Factor Theorem o.e. find quadratic factor solve cubic equation reject $t = -2$ and -5	B1 M1 f.t. B1 . M1 f.t. M1 f.t. A1

- (b) Find the total distance travelled by the particle in the interval $0 \leq t \leq 3$.

[4]

Solution	Description	Marks
$s = \int -(t+2)^3 + 9t + 18 \, dt$ $= -\frac{(t+2)^4}{4} + 9\left(\frac{t^2}{2}\right) + 18t + c_1$ $= -\frac{1}{4}(t+2)^4 + \frac{9}{2}t^2 + 18t + c_1, \text{ where } c_1 \text{ is a constant.}$ When $t = 0, s = 0$. $0 = -\frac{1}{4}(0+2)^4 + \frac{9}{2}(0)^2 + 18(0) + c_1$ $c_1 = 4$ $\therefore s = -\frac{1}{4}(t+2)^4 + \frac{9}{2}t^2 + 18t + 4$ When $t = 0, s = 0$. When $t = 1$, $s = -\frac{1}{4}(1+2)^4 + \frac{9}{2}(1)^2 + 18(1) + 4$ $= 6\frac{1}{4}$ When $t = 3$, $s = -\frac{1}{4}(3+2)^4 + \frac{9}{2}(3)^2 + 18(3) + 4$ $= -57\frac{3}{4}$ Total dist. travelled $= 6\frac{1}{4} + 6\frac{1}{4} + 57\frac{3}{4}$ $= 70\frac{1}{4} \text{ m or } 70.25 \text{ m}$	integrate to find distance sub. $t = 0, s = 0$ to find c find s at $t = 1$ or $t = 3$	M1 f.t. M1 f.t. M1 f.t. A1

- 9 (a) Prove the identity $(\cot A - \tan A)\sin A = 2\cos A - \sec A$.

[4]

Solution	Description	Marks
$\begin{aligned} \text{LHS} &= (\cot A - \tan A)\sin A \\ &= \left(\frac{\cos A}{\sin A} - \frac{\sin A}{\cos A} \right) \sin A \\ &= \cos A - \frac{\sin^2 A}{\cos A} \\ &= \cos A - \frac{1 - \cos^2 A}{\cos A} \\ &= \cos A - \frac{1}{\cos A} + \cos A \\ &= 2\cos A - \sec A \\ &= \text{RHS (proved)} \end{aligned}$	$\tan A$ or $\cos A$ $\sin^2 A = 1 - \cos^2 A$ split fraction $\sec A = \frac{1}{\cos A}$	B1 B1 B1 B1

- (b) Solve the equation $(\cot x - \tan x)\sin x = 1$ for $0 \leq x \leq 2\pi$.

[4]

Solution	Description	Marks
$\begin{aligned} (\cot x - \tan x)\sin x &= 1 \\ 2\cos x - \sec x &= 1 \\ 2\cos x - \frac{1}{\cos x} &= 1 \\ 2\cos^2 x - \cos x - 1 &= 0 \\ (2\cos x + 1)(\cos x - 1) &= 0 \\ \cos x &= -\frac{1}{2} \quad \text{or} \quad \left(\cos x = 1 \right) \\ \text{basic } \angle &= \cos^{-1}\left(\frac{1}{2}\right) \quad \text{or} \quad \left(x = 0 \text{ or } 2\pi \right) \\ &= \frac{\pi}{3} \\ x &= \pi - \frac{\pi}{3} \quad \text{or} \quad \pi + \frac{\pi}{3} \\ &= \frac{2\pi}{3} \quad \text{or} \quad \frac{4\pi}{3} \end{aligned}$	apply (a) solve quadratic	B1 M1 f.t. A1 A1

- (c) The equation $2 \cos kx - \sec kx = -\frac{7}{2}$ has exactly four solutions for $0 \leq x \leq 2\pi$, where k is a positive integer. Find the value of k . [4]

Solution	Description	Marks
$2 \cos kx - \sec kx = -\frac{7}{2}$ $2 \cos kx - \frac{1}{\cos kx} = -\frac{7}{2}$ $4 \cos^2 kx + 7 \cos kx - 2 = 0$ $(4 \cos kx - 1)(\cos kx + 2) = 0$ $\cos kx = \frac{1}{4} \quad \text{or} \quad \cos kx = -2 \quad (\text{rej.})$ Since $\cos kx = \frac{1}{4}$ only has 2 solutions, the period for $\cos kx$ must be π. OR there are 2 solutions in 1 cycle. Thus, for the equation to have exactly 4 solutions, there need to be 2 cycles. $\therefore k = 2$.	solve quadratic must state rejection	M1 f.t. M1 f.t. M1 A1

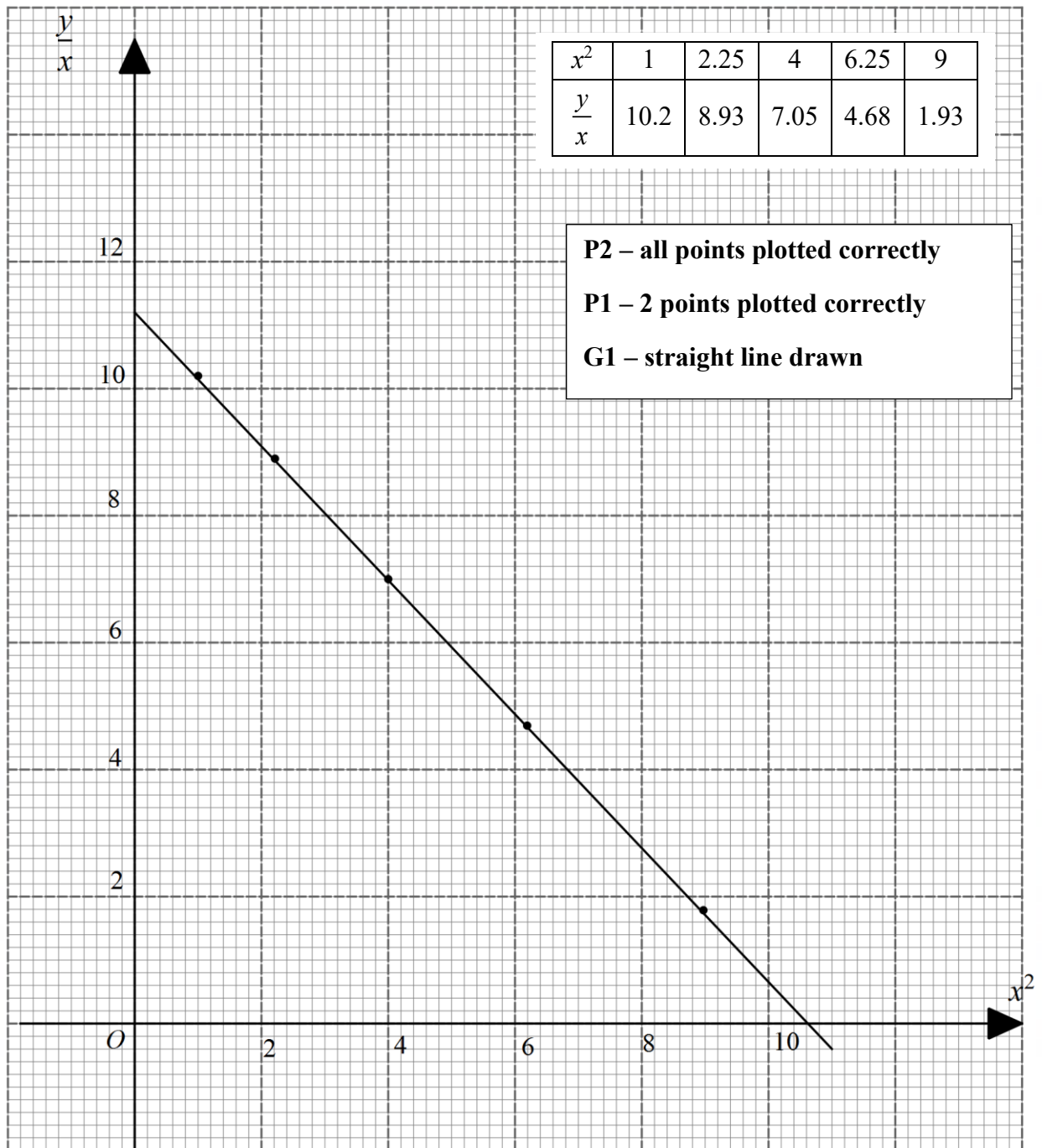
- 10 The table shows the experimental values of two variables x and y .

x	1	1.5	2	2.5	3
y	10.2	13.4	14.1	11.7	5.8

It is known that x and y are related by the equation $y = px^3 + qx$, where p and q are constants.

- (a) Plot $\frac{y}{x}$ against x^2 and draw a straight line graph.

[3]



- (b)** Use your graph to estimate the value of p and of q .

[3]

Solution	Description	Marks
$y = px^3 + qx$ $\frac{y}{x} = px^2 + q$ $\text{gradient} = \frac{11.2 - 0}{0 - 10.6}$ $p = -1.05666\dots$ $= -1.06 \text{ (3 s.f.) } \quad [\text{accept } -1.1 \text{ to } -0.9]$ $\frac{y}{x} - \text{intercept} = 11.2$ $q = 11.2 \quad [\text{accept } 11.0 \text{ to } 11.5]$	gradient formula	M1 f.t. A1 B1

- (c)** Another straight line can also be drawn from the equation $y = px^3 + qx$. Find the gradient and vertical intercept of this line.

[2]

Solution	Description	Marks
$y = px^3 + qx$ $\frac{y}{x^3} = \frac{q}{x^2} + p$ gradient = q and vertical intercept = p = 11.2 = -1.06 (3 s.f.)	manipulation	M1 f.t. A1

- 11 (a) Differentiate $\frac{x}{(1+x)^2}$ with respect to x .

[2]

Solution	Description	Marks
$\frac{d}{dx} \left[\frac{x}{(1+x)^2} \right] = \frac{(1+x)^2 (1) - x [2(1+x)(1)]}{[(1+x)^2]^2}$ $= \frac{(1+x)(1+x-2x)}{(1+x)^4}$ $= \frac{1-x}{(1+x)^3}$	numerator denominator	B1 B1

- (b) Hence, evaluate $\int_0^1 \frac{3-x}{(1+x)^3} dx$.

[4]

Solution	Description	Marks
$\int_0^1 \frac{3-x}{(1+x)^3} dx = \int_0^1 \frac{1-x}{(1+x)^3} + \frac{2}{(1+x)^3} dx$ $= \left[\frac{x}{(1+x)^2} \right]_0^1 + \left[\frac{2(1+x)^{-2}}{(-2)(1)} \right]_0^1$ $= \left[\frac{x}{(1+x)^2} - \frac{1}{(1+x)^2} \right]_0^1$ $\left[\frac{x-1}{(1+x)^2} \right]_0^1$ $= \frac{1-1}{(1+1)^2} - \frac{0-1}{(1+0)^2}$ $= 0 - (-1)$ $= 1$	split apply (a) integrate $\frac{1}{(1+x)^3}$	M1 B1 M1 f.t. A1